

Lecture – 3.1

JAVA: Introduction, Keywords, Data Types, Identifiers

Afsara Tasneem Misha

Lecturer

Department of CSE

Daffodil International University

Contents

- Introduction to Java
- Keywords in Java
- Data Types in Java
- Java Identifiers

Introduction to Java

Introduction to Java

- Java is a high-level programming language originally developed by **Sun Microsystems** and released in **1995**.

- Java runs on a variety of platforms, such as Windows, Mac OS, and the various versions of UNIX.

Features of Java

- **Object Oriented**: In Java, everything is an Object. Java can be easily extended since it is based on the Object model.
- **Platform independent:** Unlike many other programming languages including C and C++, when Java is compiled, it is not compiled into platform specific machine, rather into platform independent byte code. This byte code is distributed over the web and interpreted by virtual Machine (JVM) on whichever platform it is being run.
- **Simple:** Java is designed to be easy to learn. If you understand the basic concept of OOP Java would be easy to master.
- **Secure**: With Java's secure feature it enables to develop virus-free, tamper-free systems. Authentication techniques are based on public-key encryption.
- **Architectural-neutral:** Java compiler generates an architecture-neutral object file format which makes the compiled code to be executable on many processors, with the presence of Java runtime system.
- **Portable:** Being architectural-neutral and having no implementation dependent aspects of the specification makes Java portable. Compiler in Java is written in ANSI C with a clean portability boundary which is a POSIX subset.

Features of Java

- **Robust:** Java makes an effort to eliminate error prone situations by emphasizing mainly on compile time error checking and runtime checking.

- **Multithreaded**: With Java's multithreaded feature it is possible to write programs that can do many tasks simultaneously. This design feature allows developers to construct smoothly running interactive applications.

- **Interpreted:** Java byte code is translated on the fly to native machine instructions and is not stored anywhere. The development process is more rapid and analytical since the linking is an incremental and light weight process.

- **High Performance:** With the use of Just-In-Time compilers, Java enables high performance.

- **Distributed:** Java is designed for the distributed environment of the internet.

- **Dynamic:** Java is considered to be more dynamic than C or C++ since it is designed to adapt to an evolving environment. Java programs can carry extensive amount of run-time information that can be used to verify and resolve accesses to objects on run-time.

Tools you will need

- For performing the examples discussed in this tutorial, you will need a Pentium 200-MHz computer with a minimum of 64 MB of RAM (128 MB of RAM recommended)

You also will need the following software:

- **Operating system.**
- **JDK (Java Development Kit)**
- **Java editor**

Popular Java Editors

To write your Java programs, you will need a text editor. There are even more sophisticated IDEs available in the market. But for now, you can consider one of the following:

- **Notepad:** On Windows machine you can use any simple text editor like Notepad, TextPad.

- **Netbeans:** is a Java IDE that is open-source and free which can be downloaded from http://www.netbeans.org/index.html

- **Eclipse:** is also a Java IDE developed by the eclipse open-source community and can be downloaded from http://www.eclipse.org/

Simple Java Program

```
public class MyFirstJavaProgram {

    public static void main(String []args) {
        System.out.println("Hello World");
    }
}
```

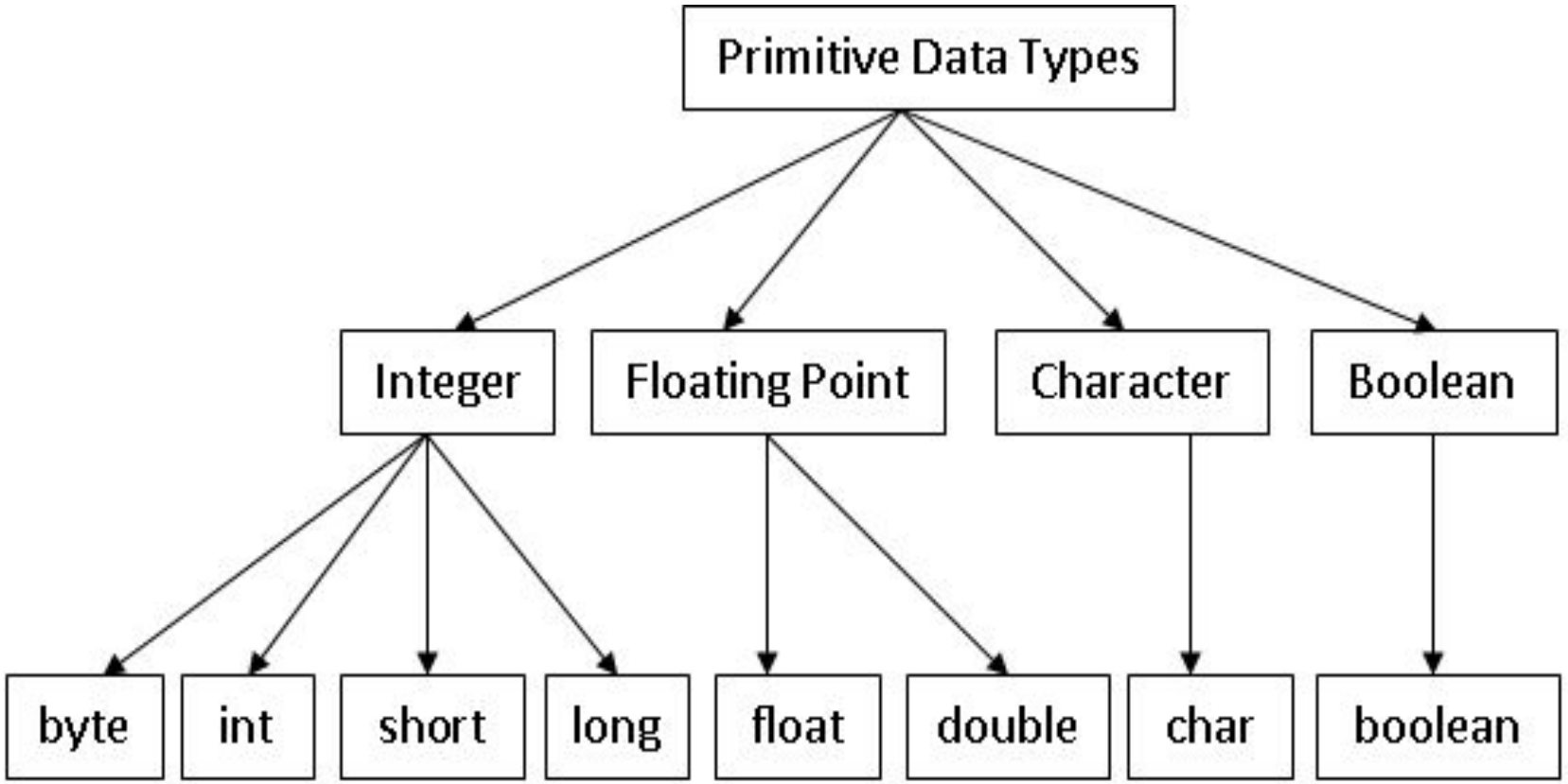
Keywords in java

Java Keywords

abstract	assert	boolean	break
byte	case	catch	char
class	const	continue	default
do	double	else	enum
extends	final	finally	float
for	goto	if	implements
import	instanceof	int	interface
long	native	new	package
private	protected	public	return
short	static	strictfp	super
switch	synchronized	this	throw
throws	transient	try	void
volatile	while		

Data Types in java

Data Types


Primitive Data Types
Integer
Floating Point
Character
Boolean
byte
int
short
long
float
double
char
boolean

PRIMITIVE DATA TYPE in JAVA

Type	Contains	Default	Size	Range
byte	Signed integer	0	8 bits	-128 to 127
short	Signed integer	0	16 bits	-32768 to 32767
int	Signed integer	0	32 bits	-2147483648 to 2147483647
float	IEEE 754 floating point	0.0f	32 bits	±1.4E-45 to ±3.4028235E+38
long	Signed integer	0L	64 bits	-9223372036854775808 to 9223372036854775807
double	IEEE 754 floating point	0.0d	64 bits	±4.9E-324 to ±1.7976931348623157E+308
boolean	true or false	FALSE	1 bit	NA
char	Unicode character	'\u0000'	16 bits	\u0000 to \uFFFF

Exercise

- **What are the differences between the constants 7, ‘7’ and “7”?**

Identifiers in java

Java Identifiers

- Names used for classes, variables and methods are called identifiers.
- All identifiers must follow the following rules:

  - An identifier is a sequence of characters that consists of **letter (A to Z or a to z), digits(0 to 9), currency character ($)** or an **underscore (_).**
  - All identifiers should begin with a letter (A to Z or a to z), currency character ($) or an underscore (_).**It can not start with a digit (0-9).**
  - A **keyword** cannot be used as an identifier.
  - Most importantly identifiers are **case sensitive**.

    area, Area, AREA are all different identifiers

Examples of legal identifiers: age, $salary, _value, __1_value

Examples of illegal identifiers: 123abc, -salary

Exercise

- **Find out the legal Identifiers**

$2 2A	area	d+4	Max-Number	
Min_number	aaa	sales_tax	_circleArea	a2b
Box100width	/$directory	ab1234$$	ComputeArea	

Thank you!

